

Powering AI in a Climate-Stressed World

Energy, Infrastructure and Climate Resilience · 19 May
2026 · Audience: strategic, infrastructure and investment
leadership
Read estimate: 3 min Executive Synthesis · 20 min full read

Executive Synthesis

The energy transition has acquired three new constraints in the past six months, and they are not arriving on the same clock. The first is the AI load curve: the International Energy Agency's base case now projects global data centre electricity demand at around 945 TWh by 2030, more than doubling the 2024 baseline of 415 TWh, with AI as the dominant driver ([IEA, Apr 2025](#)). The second is the transmission deficit: roughly 10,300 projects representing 1,400 GW of generation and 890 GW of storage were sitting in US interconnection queues at end-2024, with the typical project taking 55 months from request to commercial operation ([Lawrence Berkeley National Laboratory, Dec 2025](#)). The third is the insurability frontier: 2025 was the sixth consecutive year insured natural catastrophe losses exceeded \$100 billion, with Swiss Re putting the total at \$107 billion and the Los Angeles wildfires alone contributing \$40 billion, the largest insured wildfire event on sigma records ([Swiss Re Institute, Mar 2026](#)).

This cycle's evidence supports three structural shifts. First, the cost of AI compute is being internalised into the cost of capital across capacity markets: PJM's 2027/2028 auction cleared at the \$333.44/MW-day price cap, the third record auction in succession, with data centres responsible for around 40% of total auction costs by the market monitor's reckoning ([Utility Dive, Dec 2025](#); [Utility Dive, Dec 2025](#)). Second, the regulatory toolkit for moving electrons has finally arrived (FERC Order 1920 in May 2024, the European Commission's Affordable Energy Action Plan in February 2025, ENTSO-E's TYNDP 2026 draft portfolio of 178 transmission and 49 storage projects) but every clock on the toolkit runs in years, while the load arrives in months ([FERC, May 2024](#); [European Commission, Feb 2025](#); [ENTSO-E, Oct 2025](#)). Third, climate physical risk is moving from the prudential periphery to the credit and insurance core: California's

FAIR Plan grew to nearly \$700 billion of insured property by September 2025, a 317% increase since 2021, and First Street estimates climate risks could erase \$1.4 trillion of US property value over 30 years ([California Department of Insurance](#), Feb 2026; [First Street](#), Feb 2025).

The energy transition has acquired three new constraints in the past six months, and they are not arriving on the same clock.

For the reader, the consequence is that energy, infrastructure and climate cannot be planned as separate workstreams in the next twelve months. The Talen-AWS Susquehanna deal (1.92 GW, \$18 billion, through 2042) and Constellation's Microsoft Three Mile Island restart (835 MW, around \$1.6 billion, online by 2028) are not three isolated nuclear stories: they are evidence that hyperscalers are bypassing both the queue and the climate-risk re-pricing by buying behind-the-meter firm clean power outright ([POWER Magazine](#), Jun 2025; [Constellation](#), Sep 2024). This report adopts a posture of *integrated capital planning*: every capital decision must now be tested against compute demand, grid interconnection timing, climate physical-loss repricing, and firm-clean supply availability in a single matrix, not four. The reader who does this will pay less to connect, less to insure, and less to source clean firm power than the reader who does not.

Where this analysis could be wrong

The central assumption is that the four constraints (AI compute, transmission, climate physical loss, firm-clean supply) tighten together over the next 18 months rather than resolving independently. If hyperscaler capex moderates materially before end-2026, if FERC Order 1920 and PJM's behind-the-meter colocation reforms accelerate queue clearance rates by 30% or more inside two years, and if the 2026 Atlantic hurricane season and Western US fire season deliver below-average insured losses, the integrated planning posture is overstated and the four can be resolved sequentially. The single observable that would force a material revision is hyperscaler capex guidance through Q3 2026 earnings: a sustained 15%+ guidance reduction across Microsoft, Amazon, Google and Meta would be the disconfirming signal that the demand curve is breaking.

The decisions that cannot be deferred this cycle

1. **Integrate the four constraints into a single capital matrix before the FY27 plan locks.** Compute, queue, climate physical loss and firm-clean supply now move together; planning them in four separate workstreams is the binding strategic error of the cycle.
2. **Take a documented behind-the-meter and colocation position before PJM's reforms land.** The FERC and PJM colocation rulemaking is moving (PJM filed proposed changes 23 February 2026); waiting until the rules finalise concedes the optionality to hyperscalers and IPPs that have already pre-positioned.
3. **Reprice climate physical exposure into the cost of capital, not just the cost of insurance.** The California FAIR Plan growth, Florida Citizens depopulation, and the Bank of England 2026 DyGIST stress test all converge on the same supervisory signal: climate physical loss has moved into the prudential frame.
4. **Set the firm-clean offtake posture before the 2026 PPA window closes.** The Talen-AWS, Constellation-Microsoft, Google-Kairos-TVA and Meta nuclear deals between September 2024 and December 2025 took the available existing and restartable capacity off the market; the next window opens around advanced reactor first-of-a-kind deployments in 2028 to 2030.

Each is developed below, with a decision posture, in the four Strategic Implications.

Risk-Axis Snapshots

Four risk lenses on the same evidence base. Each card reads across all four clusters and frames the central operational risk through that lens.

HEAT & DROUGHT

Will summer cooling demand and hydropower drought collide with AI peak load?

The shift: 2025 was around 1.43C above pre-industrial, with WMO putting the five-year mean 2025-2029 above 1.5C at 70% probability, up from 47% ([World Meteorological Organization](#), May 2025). The US, China and Europe carry 85% of global data centre load, and AI training and inference peaks now coincide with peak summer cooling ([IEA](#), Apr 2025).

The question to brief: in the worst summer week of 2026, can the grid cool the data-centre fleet, run residential and commercial AC, and absorb European and western-US hydro reductions simultaneously? The PJM capacity-market price signal says it cannot, and the price is already in the auction ([Utility Dive](#), Dec 2025).

FLOOD & STORM

Where is the insurability frontier already inside the asset perimeter?

The shift: sixth consecutive \$100bn+ insured cat year, with \$40 billion from LA wildfires (the largest single wildfire event on Swiss Re sigma records) and \$51 billion of severe convective storm loss ([Swiss Re Institute](#), Mar 2026). Florida Citizens depopulated 585,432 policies in 2025 and the California FAIR Plan reached ~\$700 billion of insured property in September 2025 ([Artemis](#), Mar 2026; [California Department of Insurance](#), Feb 2026).

The question to brief: for every site, what is the residual-market exposure, reinsurance availability and rebuild-cost trajectory if the next storm or fire materialises? First Street's \$1.4 trillion 30-year property-value erosion is the baseline, not the tail ([First Street](#), Feb 2025).

GRID & CYBER

How much of the FY27 plan depends on a queue position that does not exist yet?

The shift: 1,400 GW of generation and 890 GW of storage in US queues at end-2024; 55-month typical clear time; only 13% of 2000-2019 capacity-requests reached commercial operations by end-2024 ([LBNL](#), Dec 2025). FERC Order 1920 began to bite mid-2025 and ENTSO-E TYNDP 2026 adds 178 transmission and 49 storage projects, but supply chains are tight: Wood Mackenzie projects 195% gas-turbine price escalation by 2027 with 18-36 month lead times ([Wood Mackenzie](#), Apr 2026).

The question to brief: for every load addition in the FY27 plan, what is the queue position, alternative interconnection path and supply-chain contingency? RTE's fast-track regime (400 to 1000 MW connections in three to four years) is the European policy frontier ([RTE](#), Sep 2025).

SUPPLY-CHAIN & GEOPOLITICAL

Does the FY27 capex plan account for CBAM, equipment shortages and the firm-clean PPA window?

The shift: EU CBAM compliance began 1 January 2026, first declarations due 30 September 2027, covering cement, steel, aluminium, fertiliser, electricity and hydrogen imports. US data-centre electrical equipment is forecast to triple from \$20 billion to \$65 billion by 2030, with ~600 GW of projects searching for power against 183 GW with signed supply agreements ([Wood Mackenzie](#), Mar 2026).

The question to brief: across the FY27 build, has the company priced CBAM exposure on imported intermediates, supply-chain stress on transformers, switchgear and turbines, and the closed existing-nuclear PPA window? Meta's 6.6 GW nuclear basket through 2035 says that window is now narrow ([Latitude Media](#), Jan 2026).

Signal Clusters

The cycle's signals are organised into four clusters, ranked by impact on energy, infrastructure and climate-resilience decisions. **Immediate:** changes the FY27 capital plan, the corporate structure or the proposition. **Near-Term:** changes competitive position over the next twelve months. **Longer-Range:** a multi-year structural factor to track and revisit each cycle.

1. AI compute as the new load curve

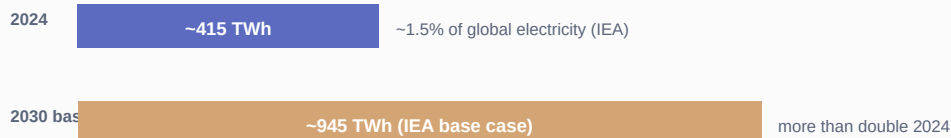
IMMEDIATE

AI compute is rewriting the shape of electricity demand in the window the grid is least able to absorb it ([IEA](#), Apr 2025). Capacity-market prices, transmission queue economics and behind-the-meter colocation rules are now one question in four jurisdictional dialects, and pull on

Clusters 2, 3 and 4 simultaneously.

The AI load curve, 2024 to 2030: doubling, with regional skew

Global data centre electricity demand: 2024 baseline to 2030 base case



2024 regional split (IEA):

United States 45% · China 25% · Europe 15% · rest of world 15%

US sector-specific scenario range (EPRI, 2024):

4.6% to 9.1% of US electricity generation by 2030; growth 3.7% to 15% annually

Source: IEA Energy and AI special report (Apr 2025) base case; EPRI Powering Intelligence white paper (May 2024) US sector-specific scenario range.

- IEA base case: global data centre demand at ~945 TWh by 2030 versus 415 TWh in 2024; US 45%, China 25%, Europe 15% of 2024 load; AI the dominant driver ([IEA Energy and AI](#), Apr 2025).
- EPRI's US sector range: data centres at 4.6% to 9.1% of US generation by 2030, 3.7% to 15% annual growth; 80% of 2023 US load in 15 states led by Virginia and Texas ([EPRI Powering Intelligence](#), May 2024).
- PJM 2027/2028 Base Residual Auction cleared at the \$333.44/MW-day cap (third record in a row); data centres drove \$6.5 billion of \$16.4 billion in auction costs (40%) per the market monitor ([Utility Dive on PJM auction](#), Dec 2025; [Utility Dive on market monitor](#), Dec 2025).
- Wood Mackenzie: US data-centre capacity from ~24 GW to 110 GW between 2026 and 2030 (68% of US load growth); ~600 GW seeking power against 183 GW with signed agreements; gas-turbine prices to rise 195% by 2027 with 18 to 36 month lead times ([Wood Mackenzie on capacity](#), Mar 2026; [Wood Mackenzie on gas turbines](#), Apr 2026).

RTE's French fast-track regime: a regulator pre-pricing AI compute

France's TSO announced the first four fast-track data-centre sites at Choose France on 19 May 2025, offering 400 to 1000 MW connections in three to four years, and from 1 August 2025 can reduce connection capacity if actual usage falls below the agreed level ([RTE](#), Sep 2025). We read this as European regulators compressing queue timing for geographically anchored compute load in exchange for usage discipline. The European Commission's Affordable Energy Action Plan reinforces the same pressure: €260 billion in projected annual savings by 2040, paired with a pricing signal designed to keep industrial competitiveness inside the EU ([European Commission](#), Feb 2025).

Counter-argument

The IEA's 945 TWh is a base case, not a forecast; EPRI's 4.6% to 9.1% range has its lower bound inside existing efficiency trajectories. IEEFA reads data centres as responsible for 63% of PJM 2025/2026 capacity-price increases (~\$9.3 billion); if FERC and PJM colocation rules push that cost back onto hyperscalers, demand growth could compress (IEEFA, Jul 2025). Sustained 15%+ hyperscaler capex moderation through Q3 2026 would be the decisive break.

This reads as accelerating only if hyperscaler capex guidance through Q3 2026 holds at or above current trajectory and if PJM-equivalent capacity-market prices in CAISO, ERCOT and ISO-NE follow PJM upward. If hyperscaler capex moderates 15%+ and capacity-market prices in non-PJM ISOs hold flat or decline through 2026, the cluster is over-stated.

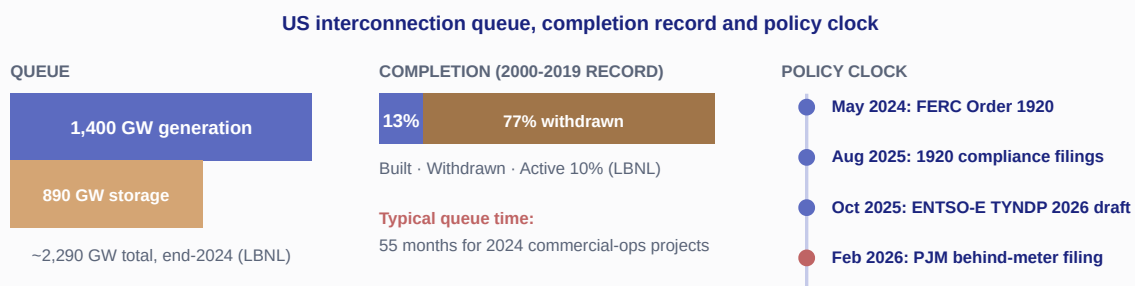
Decision link: Strategic Implications 1 (integrate the capital matrix) and 2 (behind-the-meter and colocation posture).

2. Interconnection queues and the transmission deficit

IMMEDIATE

The grid that must evacuate the new load is the binding constraint on every other bet in the cycle (LBNL [Queued Up 2025 Edition](#), Dec 2025). Any capacity plan assuming new interconnection inside two years carries a discounted probability of delivery: the regulatory tools (FERC Order 1920, ENTSO-E TYNDP 2026, RTE fast-track, PJM behind-the-meter colocation) run on multi-year clocks while demand arrives in months.

The transmission deficit: queue size, completion rate, and the policy clock



Sources: LBNL [Queued Up 2025 Edition](#) (Dec 2025); FERC Order 1920 explainer (May 2024); ENTSO-E (Oct 2025); [Utility Dive on PJM](#) (Feb 2026).

- At end-2024, ~10,300 projects (1,400 GW generation, 890 GW storage) sought US interconnection; active capacity fell 12% YoY on a historic withdrawal wave (700+ GW exited against ~500 GW new submissions); typical 2024 commercial-ops project spent 55 months in queue ([LBNL Queued Up 2025](#), Dec 2025).
- FERC Order 1920 (13 May 2024) requires regional planning every five years over a 20-year horizon; compliance filings due June and August 2025; Orders 1920-A and 1920-B clarified state involvement and cost allocation ([FERC explainer](#), May 2024).
- Niskanen reads SunZia's 17-year permit history as evidence that coordination and resource constraints, not NEPA itself, drive transmission delay ([Niskanen Center](#), Mar 2026).
- ENTSO-E's TYNDP 2026 draft adds 178 transmission and 49 storage projects (108 GW cross-border beneficial by 2040, 224 GW with 540 GW storage by 2050); PJM proposed BTMG rule changes on 23 February 2026 to facilitate data-centre colocation ([ENTSO-E TYNDP 2026](#), Oct 2025; [Utility Dive](#), Feb 2026).

RMI's "Power Couples": clean repowering as queue-avoidance

RMI estimates clean repowering (pairing large consumers with renewables near existing gas plants) could deliver 250 GW without transmission upgrades, with average savings of \$12.7 billion annually for ten years ([RMI](#), Nov 2025). We read this as evidence that the queue is not the only path to firm capacity, with the regulatory frontier (PJM's February 2026 BTMG filing, RTE's fast-track regime) starting to formalise the alternative. BloombergNEF records a record \$2.3 trillion of energy-transition investment in 2025 (up 8%, with annual growth decelerating from 27% in 2021), against a base-case Economic Transition Scenario that requires \$2.9 trillion of average annual investment over the next five years ([BloombergNEF](#), Jan 2026): a binding capital-deployment problem, not a permitting problem alone.

Counter-argument

The queue-deficit framing may overstate the binding constraint. Ember records wind and solar at 30% of EU electricity in 2025, overtaking fossil power for the first time; IRENA records nearly 700 GW of global renewable capacity added in 2025 on top of 2024's 585 GW record ([Ember](#), Jan 2026; [IRENA](#), Apr 2026). If Order 1920 compliance accelerates clearance through 2027 and European TSO delivery holds, the queue softens to a coordination issue. Clearance rate is the observable.

This reads as accelerating only if FERC Order 1920 implementation and PJM colocation rules fail to lift queue-clearance rates measurably through 2027 and if European TYNDP project delivery slips against the TYNDP 2024 baseline. If clearance rates rise above 25% within two years and TYNDP delivery holds, the cluster is over-stated.

Decision link: Strategic Implications 1 (integrate the capital matrix) and 2 (behind-the-meter and colocation posture).

3. Climate loss repricing and the insurability frontier

IMMEDIATE

Climate physical risk has moved out of the prudential periphery into the credit and insurance core this cycle ([Swiss Re Institute](#), Mar 2026; [Munich Re](#), Jan 2026). Every siting decision, capex sequencing call and long-dated project now carries an insurability and prudential exposure that did not sit on the FY24 balance sheet.

- 2025 insured cat losses at ~\$107 billion; LA wildfires the largest single insured wildfire on sigma records at \$40 billion; SCS losses \$51 billion (third-costliest year); North American wildfire losses growing 14% annually ([Swiss Re Institute](#), Mar 2026); Munich Re records \$108 billion insured against \$224 billion total damage ([Munich Re](#), Jan 2026).
- First Street estimates climate risks could cut US real-estate values by ~\$1.4 trillion over 30 years; insurance now drives a higher share of mortgage cost than ever ([First Street](#), Feb 2025).
- California's FAIR Plan reached ~\$700 billion insured property by September 2025 (52% YoY, 317% above 2021); \$1 billion assessment approved February 2025 (first in 30+ years); AB 1680 targets transparency and coverage reform ([California Department of Insurance](#), Feb 2026; [Office of the Governor of California](#), May 2026).
- Florida Citizens depopulated 585,432 policies (~\$235.6 billion exposure) in 2025, 67% below 2023 peak; Citizens estimates \$3 billion reinsurance need for the 2026 season ([Artemis](#), Mar 2026); HB 837 and forced-depopulation HB 1611 reshape the market ([Inside Climate News](#), Sep 2025).

The Bank of England 2026 DyGIST: prudential supervision catches up

The PRA's 2026 Dynamic General Insurance Stress Test (DyGIST) commences May 2026, covering insurers representing more than 80% of the PRA-regulated general insurance market by GWP ([Bank of England PRA](#), Jul 2025). PS25/25 (December 2025) and SS5/25 update SS3/19 to formalise climate-related risk expectations across UK banks and insurers. We read this as supervisory architecture moving from voluntary disclosure into binding stress-test territory inside 18 months, the observable for how quickly climate physical loss enters bank capital ratios and insurance solvency. With the Federal Reserve, OCC and FDIC having exited the Network for Greening the Financial System in early 2025, the UK clock runs materially faster than the US one on this axis.

Counter-argument

The framing may overstate the speed at which climate physical loss flows into asset prices and credit. A sixth \$100 billion+ year reads as a settled regime more than a sudden shift; the market has responded by repricing, depopulating residual markets and tightening underwriting rather than failing structurally. UNEP records full-NDC warming at 2.3-2.5C and current policies at 2.8C: the trajectory is real but credit-market repricing may be multi-decadal ([UNEP](#), Nov 2025). A below-average 2026 cat season would compress the signal.

This reads as accelerating only if the 2026 Atlantic hurricane season and Western US fire season produce insured losses at or above the 2025 baseline, if the California FAIR Plan exposure trajectory continues upward, and if the Bank of England 2026 DyGIST results require visible insurer capital top-ups. If the 2026 cat year falls 30% below 2025, the cluster is over-stated.

Decision link: Strategic Implications 1 (integrate the capital matrix) and 3 (climate physical exposure into capital).

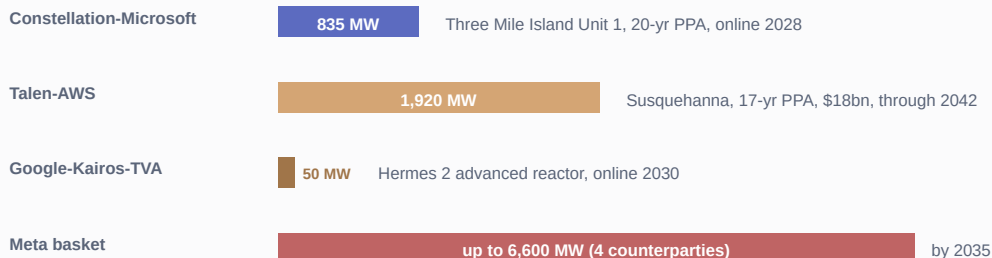
4. Hyperscaler nuclear and the firm-clean race

NEAR-TERM

Hyperscalers are bypassing the queue and the climate repricing simultaneously by buying behind-the-meter and grid-connected nuclear at scale. Between September 2024 and December 2025 the existing US nuclear fleet and its restartable units were largely absorbed into long-dated PPAs: Constellation-Microsoft, Talen-AWS, Google-Kairos-TVA and Meta's basket with Constellation, Vistra, Oklo and TerraPower ([Constellation](#), Sep 2024; [POWER Magazine](#), Jun 2025; [ANS](#), Aug 2025; [Latitude Media](#), Jan 2026). The existing-nuclear and restartable PPA window is largely closed; the next opens around SMR FOAK deployments in 2028 to 2030, with a 15 to 25% firm-clean premium above grid-constrained alternatives.

The hyperscaler nuclear basket: 2024 to 2035 firm-clean deal flow

Hyperscaler nuclear basket: deal scale, term and online date



Sources: Constellation Sep 2024, POWER Mag Jun 2025, ANS Aug 2025, Latitude Media Jan 2026

The four largest publicly-disclosed hyperscaler nuclear arrangements as of May 2026, by contracted capacity. Bars are not to a single scale but show relative deal size.

- Talen-Amazon expanded PPA (11 June 2025): up to 1.92 GW from Susquehanna through 2042; 17-year, \$18 billion; co-located SMR and uprates contemplated ([Talen Energy](#), Jun 2025; [POWER Magazine](#), Jun 2025).
- Constellation-Microsoft 20-year PPA (20 September 2024) restarts TMI Unit 1 as the Crane Clean Energy Center: ~835 MW by 2028, ~\$1.6 billion investment ([Constellation](#), Sep 2024; [Utility Dive](#), Sep 2024).
- Kairos-Google-TVA (19 August 2025): first US utility-signed PPA with an advanced reactor; 50 MW Hermes 2 online 2030 ([American Nuclear Society](#), Aug 2025); Meta added Vistra, Oklo and TerraPower agreements for up to 6.6 GW by 2035 plus a June 2025 Constellation 1.1 GW deal ([Latitude Media](#), Jan 2026).
- DOE Liftoff projects US nuclear tripling from ~100 GW (2024) to ~300 GW by 2050; 41 sites could host 60 GW new large light-water, 95 GW with advanced reactors ([US Department of Energy](#), Sep 2024); Lazard finds renewables most cost-competitive new-build unsubsidised, total LCOE rises with firming ([Lazard LCOE+](#), Jun 2025).

Hyperscaler nuclear: the Talen-AWS Susquehanna 17-year PPA

The Talen-Amazon expanded PPA is the cycle's clearest anchor on the grid-connected IPP model for hyperscaler firm-clean supply: up to 1.92 GW of carbon-free nuclear through 2042 with extension options, ramping to full volume by 2032 and potentially earlier ([Talen Energy](#), Jun 2025). The \$18 billion 17-year deal supports AI and cloud workloads at the campus adjacent to Susquehanna with delivery flexibility across Pennsylvania, and contemplates SMR build and uprates on the same site. We read the implication as a capacity-tier pricing shift: a 15 to 25% lease premium for grid-connected nuclear-backed sites is now documented in trade-press analysis ([Latitude Media](#), Apr 2026), and the existing-fleet PPA window is largely closed. The next window opens around SMR FOAK deployments in 2028 to 2030.

Counter-argument

The "firm-clean race" framing may overstate operational substitution. Hyperscaler nuclear PPAs are not yet displacing meaningful gas-fired generation; on-site gas backing hyperscaler load grew from 5% of US methane-gas power demand at end-2024 to 39% within a year, with no agreed bridge end-date ([Latitude Media on hyperscaler gas](#), Apr 2026). Off-grid campuses remain a minority per developer surveys; FERC and PJM colocation perimeters remain unsettled ([Latitude Media](#), Apr 2026). If SMR FOAK slips past 2030, firm-clean concedes another five years to gas.

This reads as accelerating only if the announced advanced-reactor first-of-a-kind deployments (Kairos Hermes 2, Oklo, TerraPower) hold to their 2030 online dates and if PJM and FERC formalise behind-the-meter colocation rules that price firm-clean preferentially. If SMR FOAK deployment slips past 2030 and hyperscaler gas-bridge growth accelerates through 2026, the cluster is over-stated.

Decision link: Strategic Implications 2 (behind-the-meter and colocation posture) and 4 (firm-clean offtake posture).

Strategic Implications

Four decisions turn this cycle's signals into the FY27 capital plan. Each names a concrete action with a horizon and a decision posture. Opportunities are expressed here as Prepare-posture moves, not as a separate section.

SI 1: Integrate the four constraints into a single FY27 capital matrix

By the Board Capital Committee meeting in Q3 2026, the Chief Strategy Officer (with the CFO and Chief Risk Officer) should produce an integrated capital matrix that tests every material capex line against four axes simultaneously: AI compute load exposure, interconnection queue position, climate physical-loss exposure and firm-clean supply availability. The matrix should be built off the IEA Energy and AI base case (945 TWh data centre demand by 2030), the LBNL Queued Up 2025 generation and storage queue, the Swiss Re and First Street climate-loss anchors, and the publicly-disclosed hyperscaler nuclear PPA list. The cost of getting this wrong sits in three concurrent failure modes: paying capacity-market prices set by AI demand, sitting in a queue that does not clear, and absorbing climate physical-loss exposure that no longer has a residual market.

Action: integrated capital matrix to Board Capital Committee by end-Q3 2026; named owner for each axis; quarterly review in FY27.

DECIDE

Draws on Clusters 1, 2 and 3.

SI 2: Take a documented behind-the-meter and colocation position before the regulatory perimeter sets

By the Board Risk Committee meeting before end-Q4 2026, the Chief Strategy Officer with General Counsel should adopt a documented position on behind-the-meter and colocation strategy in each jurisdiction in which the organisation operates: PJM (the 23 February 2026 BTMG filing), CAISO and ERCOT in the US; France (RTE fast-track regime, August 2025 capacity-reduction provision) and the EU more broadly; the UK; and Japan (METI's energy efficiency act extension to all data centres by April 2026). The window in which a pre-positioning move secures preferential queue treatment is closing as the regulators move from RFI to rule. The opportunity sits in the small set of organisations that can pre-commit to a usage-discipline regime and secure connection rights ahead of the cliff.

Action: documented jurisdictional position to Board Risk Committee by end-Q4 2026; FY27 capital plan integration.

DECIDE

Draws on Clusters 2 and 4.

SI 3: Price climate physical exposure into the cost of capital, not only the cost of insurance

By the next CFO and Treasurer briefing of the Audit Committee in FY27, the organisation should explicitly reprice climate physical exposure into the cost of capital for every long-dated asset (10+ year horizon). The supervisory architecture is moving: the Bank of England 2026 DyGIST stress test commences May 2026; California's FAIR Plan and Florida Citizens depopulation reshape residual-market exposure; First Street's \$1.4 trillion 30-year property-value erosion estimate is now the baseline. The opportunity sits in proactive repricing ahead of the supervisory wave: organisations that price climate physical exposure into cost of capital in FY27 will face lower forced capital top-ups when supervisors require it in FY28 to FY30.

Action: climate-physical-loss-adjusted cost-of-capital methodology to Audit Committee by Q2 FY27; integrate into capex hurdle rates.

PREPARE

Draws on Cluster 3.

SI 4: Monitor the firm-clean PPA window and the SMR first-of-a-kind trajectory

By the Strategy Committee's standing review, the Chief Strategy Officer should establish a quarterly monitor on the firm-clean PPA window: residual existing-nuclear and restartable capacity unbought (limited after Talen-AWS, Constellation-Microsoft, Google-Kairos-TVA, Meta basket); SMR first-of-a-kind progress (Kairos Hermes 2 to 2030, Oklo, TerraPower); and behind-the-meter colocation rule progression at FERC, PJM and equivalent. The monitor should also track hyperscaler capex guidance (the most direct demand signal) and gas-turbine supply chain pricing (the firm-fossil bridge cost). Where the organisation needs firm clean power inside 2028 to 2032, the practical option set is narrow: a partial nuclear PPA tranche from existing fleet survivors, a position on a named SMR FOAK project, or a behind-the-meter renewable-plus-storage arrangement.

Action: standing firm-clean monitor to Strategy Committee quarterly from Q3 2026; named trigger events for promotion to Prepare or Decide posture.

MONITOR

Draws on Clusters 1 and 4.

Scenario Matrix

Two uncertainties will shape the next two to five years of energy, infrastructure and climate resilience more than any other. First, the AI capex curve: does hyperscaler capex remain at or above current trajectory, or does it correct materially through 2026 to 2027? Second, climate physical loss: does the 2026 to 2028 catastrophe-loss curve accelerate above the 2025 baseline, or moderate at or below the 2024 to 2025 mean? The four scenarios below are planning aids, not forecasts.

The Twin Squeeze

AI capex holds at or above trajectory through 2027; climate physical loss accelerates above the 2025 baseline. The four constraints (compute, queue, climate, firm-clean) tighten simultaneously. Capacity prices in PJM-equivalent ISOs continue to escalate past the \$333.44/MW-day 2027/2028 cap; the FAIR Plan and Florida Citizens come under fresh stress on a major hurricane or fire event; hyperscaler firm-clean PPA premiums widen. Organisations that have not pre-positioned behind-the-meter and on firm-clean supply pay both the queue premium and the climate-loss premium.

Early indicators: hyperscaler Q3 2026 capex guidance held or raised; Atlantic 2026 hurricane season above the 2025 baseline; PJM 2028/2029 BRA clears at cap; SMR FOAK projects hold 2030 schedules.

Compute Cools, Climate Hardens

AI capex corrects 15%+ through 2026 to 2027; climate physical loss continues at or above 2025 baseline. The grid and queue pressures soften, but the climate physical-loss repricing continues. Insurance and reinsurance markets concentrate, residual markets grow, and supervisory architecture (PRA DyGIST, EU climate stress-tests, FSOC climate observations) starts to feed into prudential capital. The firm-clean PPA window narrows but the urgency softens; nuclear capex sequencing extends. Organisations face climate physical-loss capital costs without the compute-driven offset on revenue.

Early indicators: hyperscaler capex guidance reduces 15%+ across at least three of four hyperscalers; 2026 Atlantic and Pacific cat year above 2025 baseline; PRA DyGIST 2026 results require visible insurer capital top-ups; FAIR Plan exposure continues upward.

Compute Wins, Climate Moderates

AI capex holds; climate physical loss moderates 30%+ below 2025 baseline. The compute load grows but the climate-physical repricing reverses temporarily, reopening insurance markets and softening prudential pressure. Capacity-market and queue-clearance constraints remain binding; firm-clean PPA premiums hold but the urgency on climate physical exposure compresses. Organisations face a clearer single-axis problem (compute and grid) with the climate axis softer for one cycle.

Early indicators: hyperscaler capex guidance held or raised; 2026 cat year 30%+ below 2025 baseline; reinsurance market softening visible in 1 January 2027 renewals; FAIR Plan exposure plateaus.

The Soft Landing

AI capex moderates 15%+; climate physical loss moderates 30%+ below 2025. The most benign scenario for incumbent grid and infrastructure planning. Capacity-market prices ease, queue clearance accelerates, climate physical-loss capital costs soften, and the firm-clean PPA window reopens without urgency. The risk is complacency: pre-positioning that locked in for the Twin Squeeze yields lower returns, but the optionality value of the positions remains.

Early indicators: hyperscaler capex moderation visible across hyperscalers; 2026 cat year and PJM capacity-market both ease; SMR FOAK schedules slip without pricing pressure; supervisory climate-stress test results modest.

What We Are Not Planning For

A near-term collapse of US federal climate policy as a binding planning shock

UNEP records that the new round of NDCs has barely moved the projected 2.3-2.5C warming pathway once the announced US Paris withdrawal is netted; the Federal Reserve, OCC and FDIC have exited the Network for Greening the Financial System; SEC climate-disclosure rules remain unsettled. We treat the absence of US federal climate policy momentum as the planning baseline, not as a shock. The supervisory frame moves instead through the Bank of England PRA, EU stress-tests, California and Florida state action and reinsurance pricing.

Reinstate if: a US federal climate-related financial risk rule is finalised inside the cycle window, or the Federal Reserve re-enters the Network for Greening the Financial System with a published timeline.

Direct hyperscaler-grid disconnects (off-grid AI campuses) as the dominant model

Trade-press analysis and developer interviews indicate off-grid data-centre campuses remain a minority of capacity; the regulatory perimeter for behind-the-meter colocation is unsettled but not closed. The deal-flow evidence (Talen-AWS at Susquehanna grid-connected; Constellation-Microsoft at TMI grid-connected; Google-Kairos-TVA on TVA grid) supports grid-connected firm-clean as the dominant pattern in this cycle. Off-grid is in the Monitor category, not Prepare or Decide.

Reinstate if: a major hyperscaler publicly commits to an off-grid AI campus of 500 MW+ with a binding online date inside 2028, or FERC issues a final rule materially preferential to off-grid colocation.

EU CBAM as an immediate energy-cost shock for the focal industries

EU CBAM entered its compliance phase 1 January 2026; first declarations due 30 September 2027; certificate sales begin 1 February 2027 with prices tied to quarterly EUA averages and an importer requirement to hold 50% of embedded emissions (down from 80%) on the registry. The mechanism applies to cement, iron and steel, aluminium, fertilisers, electricity and hydrogen imports. For most non-EU-importing organisations in scope of this report, CBAM is a Monitor item across the FY27 to FY28 planning window rather than an FY27 capital-shock event. The shock arrives for EU-importing energy-intensive industries.

Reinstate if: the organisation's CBAM-import exposure exceeds the 50-tonne threshold inside FY27, or if the CBAM certificate price tracks above 80 EUR/tonne.

A binding Atlantic carbon-market price floor or border-adjustment cascade

Beyond EU CBAM, no other major jurisdiction has finalised an equivalent carbon-border mechanism inside the cycle window. UK ETS-CBAM is in design, Canada and Australia are in discussion, the US has no federal carbon price. We treat a cascade of Atlantic carbon-border mechanisms as a multi-year structural watch item, not an FY27 planning shock.

Reinstate if: a second G7 jurisdiction (UK, Canada, Japan) finalises a binding carbon-border mechanism inside the cycle window.

Discussion Points for the Leadership Team

1. By end-Q3 2026 do we hold a single integrated capital matrix that tests every material capex line against AI compute load, queue position, climate physical-loss exposure and firm-clean supply? If not, where exactly are the four axes being planned separately, and who owns each?
2. For every major site in the FY27 plan, do we have a documented behind-the-meter and colocation position consistent with the live regulatory perimeter in that jurisdiction (PJM, CAISO, ERCOT, RTE, the EU more broadly, UK, METI in Japan)?

3. The Talen-AWS, Constellation-Microsoft, Google-Kairos-TVA and Meta deals collectively absorbed the bulk of existing US nuclear and restartable capacity inside fifteen months. Where in the FY27 to FY30 plan is our firm-clean offtake position relative to the SMR first-of-a-kind cohort, and what is the named trigger to promote it from Monitor to Prepare or Decide?
4. The California FAIR Plan is at nearly \$700 billion of insured property and the Florida Citizens depopulation has removed \$235.6 billion of state-backed exposure inside 2025. For our property portfolio, what is the residual-market exposure, the reinsurance availability and the rebuild-cost trajectory? Are we using the First Street \$1.4 trillion baseline or a softer climate-loss assumption?
5. FERC Order 1920 compliance filings landed mid-2025 and ENTSO-E's TYNDP 2026 portfolio is in draft. For each of our major load additions, is the assumed interconnection timeline calibrated to the LBNL 55-month median, or to a faster optimistic case?
6. The Bank of England 2026 DyGIST commences May 2026 and the PRA's PS25/25 / SS5/25 update climate-risk expectations. Are our UK-regulated subsidiaries (if applicable) calibrated for the DyGIST output window, and is our group-level climate stress methodology aligned?
7. The gas-turbine market faces 18-36 month lead times and Wood Mackenzie projects 195% price escalation by 2027. Where is gas in our firm-bridge plan, and what is the documented end-date or off-ramp for that bridge?
8. The IEA 945 TWh 2030 base case and the EPRI 4.6% to 9.1% US 2030 sector range bracket the AI compute load. Which scenario does our FY27 revenue and capacity model assume, and what triggers a shift between scenarios mid-cycle?
9. The European Commission's Affordable Energy Action Plan targets €260 billion in annual savings by 2040; the EU CBAM compliance phase began 1 January 2026. For European-domiciled operations, is the energy-cost trajectory calibrated to the Action Plan path or to status quo?
10. If hyperscaler capex guidance moderates 15%+ through Q3 2026 earnings (the disconfirming signal in the Executive Synthesis stress-test), what is our pre-committed response: do we accelerate, hold, or compress our own infrastructure timing?

Source Confidence Register

This cycle drew on 41 verified sources across the four clusters, captured against a soft six-month recency window. One structural anchor per cluster is permitted in the 6-12 month band and is marked as such in the tables below. Tier mix: 10 Tier 1, 16 Tier 2, 13 Tier 3, 2 Tier 4 (63% Tier 1 or 2). Language mix: 39 English, 1 French, 1 Japanese. The evidence concentrates geographically where the signal does: US-heavy on the AI-compute and interconnection clusters, EU and UK on the climate-prudential and grid clusters. The thinnest evidence axis is climate-physical-loss feed-through into commercial-real-estate credit; we read this as a coming-cycle topic and flag it for next-cycle expansion.

Source tiers: **Tier 1**, governments, regulators and intergovernmental bodies. **Tier 2**, think-tanks, academic institutes, major consultancies and quality data providers. **Tier 3**, quality journalism and specialist trade press. **Tier 4**, vendor, company and practitioner sources, used only as directional corroboration.

Cluster 1: AI compute as the new load curve

Source	Tier	Date	Key claim used
IEA, Energy and AI World Energy Outlook Special Report	Tier 1	Apr 2025 (structural anchor)	Data centre electricity consumption projected to reach around 945 TWh by 2030, more than doubling 2024 baseline of 415 TWh; AI the dominant driver; US 45%, China 25%, Europe 15% of 2024 load.
EPRI, Powering Intelligence	Tier 2	May 2024 (structural anchor)	US data centres at 4.6% to 9.1% of US electricity generation by 2030; growth 3.7% to 15% annually; 80% of 2023 US load in 15 states led by Virginia and Texas.
Wood Mackenzie, US data center capacity	Tier 2	Mar 2026	US data center capacity to grow 24 GW to 110 GW between 2026 and 2030 (68% of US load growth); 600 GW seeking capacity against 183 GW with signed agreements.
Utility Dive on PJM auction	Tier 3	Dec 2025	PJM 2027/2028 auction cleared at \$333.44/MW-day cap; 145,777 MW procured, 6,625 MW below reliability target; third record auction in a row.
Utility Dive on PJM market monitor	Tier 3	Dec 2025	Data centers were \$6.5bn of \$16.4bn (40%) PJM December auction costs per market monitor; 5,100 MW of peak load increase attributable to data centers.

Source	Tier	Date	Key claim used
IEEFA, PJM capacity prices and data centers	Tier 2	Jul 2025	PJM capacity prices rose from \$28.92 to \$329.17/MW-day from 2024/25 to 2026/27; data centers responsible for 63% of the 2025/2026 increase (\$9.3bn).
Wood Mackenzie, gas turbine prices	Tier 2	Apr 2026	Gas turbine prices projected to rise 195% by 2027; 18 to 36 month equipment lead times entrenched.
RTE, data centers in France	Tier 2	Sep 2025	French data center consumption projected at 20 TWh by 2030 and 35 TWh by 2035; fast-track regime offers 400 to 1000 MW connections in three to four years.
Wood Mackenzie, 160 GW large-load commitments	Tier 2	Oct 2025 (structural anchor)	US utility large-load commitments have reached over 160 GW, equivalent to 22% of 2024 US peak load, dominated by data centre development; PJM utilities forecast 55 GW of large-load growth by 2030 and 100 GW by 2037.
European Commission, Affordable Energy Action Plan	Tier 1	Feb 2025 (structural anchor)	EU Action Plan for Affordable Energy targets €45bn savings 2025, €130bn annually by 2030 and €260bn by 2040, paired with the Clean Industrial Deal.

Cluster 2: Interconnection queues and the transmission deficit

Source	Tier	Date	Key claim used
FERC, Order 1920 explainer	Tier 1	May 2024 (structural anchor)	Order 1920 requires long-term regional planning at least every five years looking 20 years forward; rule effective 12 Aug 2024; compliance filings June and August 2025.
LBNL, Queued Up 2025 Edition	Tier 2	Dec 2025	Approximately 10,300 projects with 1,400 GW generation and 890 GW storage in US queues at end-2024; typical 2024 commercial-ops project spent 55 months in queue; only 13% of 2000-2019 capacity-requests reached commercial operations by end-2024.

Source	Tier	Date	Key claim used
Niskanen Center, transmission gap	Tier 2	Mar 2026	SunZia transmission line took 17 years to permit; coordination and resource constraints rather than NEPA itself are the primary delay driver; need for high-voltage planning parity, clean-firm solutions, NEPA updates.
ENTSO-E, TYNDP 2026 draft portfolio	Tier 1	Oct 2025	TYNDP 2026 draft contains 178 transmission and 49 storage projects; 108 GW additional cross-border capacity beneficial by 2040; 224 GW with 540 GW storage by 2050.
Utility Dive, PJM behind-the-meter	Tier 3	Feb 2026	PJM proposed 23 February 2026 changes to Retail Behind The Meter Generation rules to facilitate co-locating generation with data centers.
RMI, Waiting in Queue	Tier 2	Nov 2025	Clean repowering offers potential for 250 GW of new renewables without transmission upgrades, saving an average of \$12.7bn annually for ten years; Power Couples strategy.
BloombergNEF, Energy Transition Investment Trends 2026	Tier 2	Jan 2026	Global energy transition investment reached a record \$2.3tn in 2025, up 8% from 2024 (growth slowing from 27% in 2021); largest drivers electrified transport \$893bn, renewables \$690bn, grid investment \$483bn; base-case ETS path needs \$2.9tn average annual over the next five years.
Ember, European Electricity Review 2026	Tier 3	Jan 2026	Wind and solar generated a record 30% of EU electricity in 2025, surpassing fossil power for the first time; solar grew 20.1%; renewables 48% of EU power overall.
IRENA, 2025 capacity surge	Tier 2	Apr 2026	Global renewable capacity additions approached 700 GW in 2025 building on 585 GW in 2024; solar over three-quarters of expansion; still short of 16.6% annual growth required to triple by 2030.
METI Japan, Strategic Energy Plan	Tier 2	Feb 2025 (structural anchor)	Japan's Seventh Strategic Energy Plan anticipates rising electricity demand from data centres and semiconductor factories; Energy Efficiency Act extended to all data centres by April 2026.

Cluster 3: Climate loss repricing and the insurability frontier

Source	Tier	Date	Key claim used
Swiss Re Institute, 2025 insured cat losses	Tier 2	Mar 2026	2025 insured natural catastrophe losses around \$107bn (sixth consecutive year above \$100bn); LA wildfires largest single insured wildfire event on sigma records at \$40bn; severe convective storms \$51bn.
Munich Re, 2025 natural disaster figures	Tier 2	Jan 2026	\$108bn insured against \$224bn total damage 2025; climate change linked to LA wildfires and Atlantic hurricanes.
First Street, 12th National Risk Assessment	Tier 2	Feb 2025 (structural anchor)	Climate-related risks could reduce US real estate values by approximately \$1.4tn over 30 years; insurance cost now a higher share of mortgage cost than ever before.
California Department of Insurance, FAIR Plan reform	Tier 1	Feb 2026	AB 1680 (Make it FAIR Act); FAIR Plan insuring nearly \$700bn (52% YoY, 317% above 2021); \$1bn assessment Feb 2025 (first in 30+ years); State Farm share over \$165m.
Office of the Governor of California, State Farm enforcement	Tier 1	May 2026	State enforcement action against State Farm over 2025 LA wildfire claims handling; 17% homeowner rate hike approved post-fires; major insurers reducing California exposure.
Artemis, Florida Citizens entering 2026	Tier 3	Mar 2026	Florida Citizens depopulated 585,432 policies removing \$235.6bn exposure 2025; 67% lower than 2023 peak; \$3bn reinsurance / cat-bond need for 2026 season.
Bank of England PRA, 2026 DyGIST	Tier 1	Jul 2025	2026 Dynamic General Insurance Stress Test commences May 2026 covering insurers representing over 80% of PRA-regulated general insurance market by GWP; PS25/25 updates SS3/19 on climate-related risks.
Inside Climate News, Florida insurance crisis	Tier 3	Sep 2025	Florida insurance crisis hits hardest in disadvantaged counties; HB 837 tort reform (2023) and HB 1611 forced depopulation reshape market; Citizens reserves exposed to hurricane drawdown.

Source	Tier	Date	Key claim used
WMO, Global Annual to Decadal Climate Update	Tier 1	May 2025 (structural anchor)	86% chance global average temperature exceeds 1.5C in at least one of next five years; 70% chance the five-year mean for 2025-2029 exceeds 1.5C, up from 47%; 2015-2025 are the eleven hottest years on record.
UNEP, Emissions Gap Report 2025	Tier 1	Nov 2025	Full NDC implementation implies 2.3-2.5C warming; current policies 2.8C; new NDCs barely move the needle once US Paris withdrawal netted; fossil CO2 around 69% of current emissions.

Cluster 4: Hyperscaler nuclear and the firm-clean race

Source	Tier	Date	Key claim used
US Department of Energy, Pathways to Commercial Liftoff: Advanced Nuclear (Sep 2024)	Tier 1	Sep 2024 (structural anchor)	US nuclear capacity could triple from ~100 GW in 2024 to ~300 GW by 2050; 41 sites identified for 60 GW new capacity, expandable to 95 GW with smaller advanced reactors.
Talen Energy, expanded Amazon PPA	Tier 4	Jun 2025 (structural anchor)	Talen-AWS PPA: up to 1.92 GW carbon-free nuclear power from Susquehanna through 2042 with extension options; 17-year deal valued at \$18bn; ramp to full volume by no later than 2032.
POWER Magazine, Talen-Amazon analysis	Tier 3	Jun 2025 (structural anchor)	Independent confirmation of \$18bn 17-year, 1.92 GW grid-connected Susquehanna PPA; grid-connected IPP model for hyperscaler power supply.
Constellation Energy, Crane Clean Energy Center	Tier 4	Sep 2024 (structural anchor)	20-year PPA with Microsoft (20 Sep 2024) to restart Three Mile Island Unit 1 as Crane Clean Energy Center; ~835 MW carbon-free by 2028 pending approvals; ~\$1.6bn investment.
Utility Dive, TMI restart for Microsoft	Tier 3	Sep 2024 (structural anchor)	Constellation plans 2028 restart of TMI Unit 1 (Crane Clean Energy Center) under 20-year Microsoft PPA, supplying 835 MW carbon-free baseload into PJM.

Source	Tier	Date	Key claim used
American Nuclear Society, Kairos-TVA-Google PPA	Tier 3	Aug 2025	Kairos Hermes 2 to supply 50 MW to TVA grid for Google data centers AL and TN; first US utility-signed PPA with an advanced nuclear plant; online 2030.
Latitude Media, Meta nuclear basket	Tier 3	Jan 2026	Meta announces three nuclear agreements (Vistra, Oklo, TerraPower) unlocking up to 6.6 GW of clean energy by 2035; June 2025 Constellation deal for 1.1 GW.
Lazard LCOE+ June 2025	Tier 2	Jun 2025 (structural anchor)	Renewables most cost-competitive new-build generation unsubsidised against gas, nuclear, coal; onshore wind ~\$50/MWh, utility-scale solar ~\$61/MWh 2024; storage declining sharply; total LCOE rises with firming.
BloombergNEF, New Energy Outlook 2026	Tier 2	May 2026	Global data centre capacity reached 84 GW in 2025 consuming 500 TWh (1.9% of global demand), up 20% year-on-year, more than doubling to 1,114 TWh by 2050; solar becomes the world's largest electricity generator by 2032; transition reframed as an energy-security strategy.
Latitude Media, hyperscalers and gas bridge	Tier 3	Apr 2026	Hyperscalers position gas as a bridge without end-dates; on-site gas data center load grew from 5% of US methane gas power demand at end-2024 to 39% within a year.
Latitude Media, off-grid data centers	Tier 3	Apr 2026	Off-grid data centres a minority of capacity per developer surveys; grid-connected firm-clean sites carry 15-25% lease premium; FERC and PJM colocation rules unsettled.

Claim-fidelity self-disclosure

Transparent audit trail for every claim that goes beyond what a cited source literally says. Three categories per the source-discipline protocol.

1. Verbatim quotes and lists used

- "around 945 TWh by 2030" reused verbatim from the IEA Energy and AI executive summary in the report's opening paragraph and figure caption ([IEA](#), Apr 2025). No other verbatim quotes used; no ranked lists copied from sources.

2. Faithful summaries (anchoring superlatives, ranked or strong-verb claims)

- "largest insured wildfire event on sigma records" attributed to Swiss Re in the Executive Synthesis and Cluster 3 framing, where Swiss Re's press release uses substantively the same characterisation for the LA wildfires ([Swiss Re Institute](#), Mar 2026).
- "third record auction in succession" used for PJM, traceable to Utility Dive's framing of the 2027/2028 BRA against the prior two records ([Utility Dive](#), Dec 2025).
- "first US utility-signed PPA with an advanced nuclear plant" used for Google-Kairos-TVA Hermes 2, sourced from the American Nuclear Society characterisation of the August 2025 deal ([ANS](#), Aug 2025).
- "sixth consecutive year insured natural catastrophe losses exceeded \$100 billion" used for the 2025 cat year, traceable verbatim to the Swiss Re Institute framing ([Swiss Re](#), Mar 2026).
- "the typical project taking 55 months from request to commercial operation" sourced from the LBNL Queued Up 2025 Edition where it appears as the published median ([LBNL](#), Dec 2025).

3. Analyst extrapolations (claims going beyond what sources literally state, marked with analyst-voice frames)

- The Executive Synthesis claim that the four constraints "tighten together over the next 18 months rather than resolving independently" is an analyst synthesis across the four clusters' evidence; we mark this as a falsifiable claim with the named disconfirming signal (hyperscaler capex moderation 15%+ through Q3 2026) in the stress-test box.
- The cluster framings characterising "integrated capital planning" as the report's editorial posture are analyst frames, not source frames; the language is the report's, the evidence is from each cluster's sources.
- "This cycle's evidence supports three structural shifts" in the Executive Synthesis is an analyst synthesis across the IEA, PJM, FERC, BoE PRA and First Street source set; we mark it explicitly as the report's reading rather than as a sourced fact.
- The scenario-matrix axes (AI capex sustained versus corrected; climate physical loss accelerating versus moderating) are analyst constructions derived from the evidence, not lifted from any single source.
- The lead-check sentences at the end of each cluster are explicit analyst falsifiability statements, not source claims; each names a disconfirming observable.
- "The existing-fleet PPA window is largely closed" in Cluster 4 is an analyst reading across Talen-AWS, Constellation-Microsoft, Google-Kairos-TVA and Meta deal flow; no single source asserts the window's closure as such.

- The "15 to 25% lease premium for grid-connected nuclear-backed sites" referenced in Cluster 4's framing is taken from Latitude Media's reporting of current market transactions; the report's framing reads it as a settled premium, which is a slight analyst strengthening of the source's "current market transactions" qualifier ([Latitude Media](#), Apr 2026).

Cycle-over-cycle continuity claims: none. This is the inaugural cycle on this topic; no prior memory.json exists to cross-reference. The Executive Synthesis explicitly establishes this cycle as the baseline.